

engineering institutes, the national laboratories and medical schools break existing barriers, and brainstorm and work together at the interface of disciplines and innovate. While we have created wealth through outsourcing, a lot more wealth can be tapped through innovation within India that has global impact.

Three ingredients are key to innovation—freedom, funding and people. Our academic institutions have traditionally been under the purview of the Government. While some departments have enabled freedom and facilitated innovation, others have exerted stifling control that can destroy the very ethos of innovation. For our universities and national institutes to emerge as global leaders, it is vitally important that the government realise its role as a facilitator to create an environment where innovation is rewarded but also ensures the translation of the innovation to benefit the society. Many recent steps taken by the Government of India have been in the right direction. For example, the Department of Biotechnology is setting up a translational health sciences and technology institute (THSTI) that will integrate medicine, engineering and basic sciences, and business. Similarly, diversification within the IITs is a progressive step. However, this push for research needs to happen in our universities to broadly seed the culture of creative thinking and innovation during the formative stages of a student.

To create an innovation culture we need outstanding people. We need to completely overhaul the existing system—it is important to realise that recruiting faculty is not about just filling up a position but attracting the very best to be competitive. We need to urgently realise that academia is in competition with the corporate world to fish in the same talent pool. Similarly, given the urgent need for faculty and researchers, we need to conduct international searches without any nationality barrier—the only criteria being excellence. This will mean that we need to ask ourselves what package we can offer to lure the top brains from any top tier global institution, which offer start-up packages exceeding \$1m. The first step will be to move away from a system where salary is pegged to a certain government scale. Our institutions need funds to facilitate attractive start-up packages and creative salary structures. This is where the society needs to step in—most professorships in

top tier international universities are named chairs endowed by philanthropists. We have a philanthropic culture; we donate millions to temples. Interestingly, if one looks back in history, our temples were the abodes of learning. We need to realise that our academic institutions are the modern temples of learning, and need philanthropy to create institutions that compete with Harvard or Stanford.

Similarly, public-private partnerships should be actively encouraged, and could potentially mean that the best brains could be recruited on corporate salaries but enjoy the flexibility and freedom of academia. This will create an environment that will attract the best and foster innovation. Such significant funding commitments will also demand greater accountability from the system. There is an urgent need for a tenure track process, where the faculty is objectively evaluated for its performance

TO CREATE A CULTURE OF INNOVATION, WE NEED OUTSTANDING PEOPLE AND HAVE TO OVERHAUL THE SYSTEM.

by peers, and non-performers are weeded out.

While we have made progress in our healthcare status over the last two decades, we still lag behind the developed world. For example, according to World Health Organization statistics, the adult mortality rate (probability of dying between the ages of 15 and 60 per 1,000 population) in the US in 1990 and 2006 was 136 and 109 respectively, while that in the UK was 104 and 80. In comparison the adult mortality rate of India in 1990 was 278 and 241 in 2006. Even Brazil has a better record than us, with an adult mortality rate of 212 in 1990 and 176 in 2006. Clearly, as a nation, we need to strive for health status descriptors that are comparable to the developed world. This will require increased resources but with greater accountability, restructuring of existing healthcare delivery mechanisms and improved preventive and social support systems to address both chronic and acute pathologies as well as emerging lifestyle diseases. But this alone will not be sufficient in a complex country like India—we need to think out of the box—merge medicine with our strengths in technology to improve accessibility and affordability.

To leapfrog we need radical solutions. Yes, addressing core issues is vitally important, but many times innovation can offer the best answer.

Healthcare, which is a \$35-billion industry in India, is expected to reach \$75 billion by 2012 and \$150 billion by 2017.

The clinical research industry has grown from a paltry Rs 20 crore in 2001 to touch Rs 5,000 crore by 2010.



Shiladitya Sengupta, assistant professor, medicine, Brigham and Women's Hospital, Harvard Medical School, is the co-founder of Cerulean Pharmaceuticals.